

PROVIDENIYA BAY, Russian Federation

WMO: 255940

Lat: 64.378N

Lon: 173.243W

Elev: 72

StdP: 14.66

Time Zone: 12.00 (E12)

Period: 90-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-18.7	-28.0	1.2	-21.9	-24.3	1.4	-17.7	41.8	13.9	37.5	11.0	2.3	350	0.918

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	59.7	51.3	56.5	49.9	53.7	48.5	52.6	57.7	50.8	54.7	49.5	52.5	10.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.0	53.7	52.9	48.6	50.7	51.5	47.6	48.9	50.6	21.8	58.4	20.8	55.0	20.0	52.8	61.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.6	29.2	24.8	-25.5	67.6	6.1	4.1	-29.9	70.5	-33.5	72.9	-36.9	75.1	-41.3	78.1		
			-25.7	55.8	5.7	2.5	-29.9	57.6	-33.2	59.0	-36.4	60.4	-40.6	62.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.3	4.3	4.7	6.5	18.7	31.7	41.7	48.0	47.2	40.6	30.2	20.2	8.5	
	DBStd	18.57	13.28	13.89	11.65	9.04	6.58	4.90	4.03	3.41	5.73	6.15	9.99	12.68	
	HDD50	9053	1416	1268	1347	940	566	255	86	99	283	613	894	1286	
	HDD65	14483	1881	1688	1812	1390	1031	698	526	553	731	1078	1344	1751	
	CDD50	45	0	0	0	0	0	7	25	11	2	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	1	0	0	0	0	0	1	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	10.1	11.8	12.4	10.3	9.6	8.4	8.0	8.0	8.9	9.5	9.7	12.2	12.5	
Precipitation	PrecAvg	26.2	1.9	1.6	1.2	1.4	1.3	1.3	2.2	3.3	2.9	3.1	3.7	2.2	
	PrecMax	44.3	4.3	7.1	3.7	5.9	5.2	4.1	5.5	7.1	9.1	7.3	8.3	5.6	
	PrecMin	14.8	0.2	0.0	0.0	0.0	0.1	0.1	0.2	0.1	0.1	0.6	0.4	0.5	
	PrecStd	7.5	1.3	1.6	0.9	1.4	1.3	1.0	1.6	1.7	2.2	1.9	2.3	1.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	34.3	33.3	37.3	50.4	63.6	66.3	61.1	56.7	44.0	39.2	37.4	
		MCWB	31.4	32.8	31.3	33.1	43.9	51.5	54.8	52.2	48.7	41.5	34.9	34.2	
	2%	DB	31.5	31.5	31.5	34.2	45.5	57.3	60.8	57.3	52.4	41.6	37.3	33.2	
		MCWB	30.2	30.5	29.8	31.8	40.2	48.4	52.4	50.9	47.8	39.6	35.1	31.7	
	5%	DB	29.3	28.7	27.7	32.9	41.8	53.3	57.5	54.1	50.1	40.1	35.7	30.7	
		MCWB	28.1	27.8	25.9	31.0	37.7	46.3	50.8	49.4	47.4	38.4	34.0	29.4	
	10%	DB	23.2	24.5	24.0	31.2	39.3	50.0	54.6	52.1	48.4	38.2	33.7	27.1	
		MCWB	21.5	23.2	22.3	29.5	36.1	44.9	49.4	48.7	46.3	36.5	32.2	25.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	33.1	31.7	33.8	44.0	52.8	56.0	54.9	50.6	42.2	37.1	35.1
			MCDB	32.5	33.7	32.9	35.7	49.7	61.6	63.9	58.3	53.7	43.3	37.9	37.0
2%		WB	30.3	30.5	29.9	32.4	40.6	49.3	53.2	52.0	49.1	40.1	35.6	32.1	
		MCDB	31.4	31.4	31.4	33.9	44.6	55.5	58.9	55.3	51.4	41.2	36.6	33.0	
5%		WB	28.0	27.7	26.1	31.5	38.2	47.3	51.3	50.4	47.9	38.5	34.1	29.4	
		MCDB	29.4	28.7	27.4	32.8	41.4	52.6	56.1	52.9	49.8	39.7	35.3	30.7	
10%		WB	21.9	23.3	22.4	29.6	36.5	44.9	49.9	49.2	46.6	36.9	32.2	25.4	
		MCDB	23.0	24.5	23.9	31.1	38.9	49.5	53.6	51.3	48.6	38.2	33.6	26.9	
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.7	10.8	10.4	7.5	9.5	8.7	7.3	7.3	6.5	7.3	8.3	
		MCDBR	9.6	10.5	9.7	9.2	10.7	17.0	15.1	12.1	7.9	7.0	6.4	10.2	
		MCWBR	9.4	10.5	9.1	8.0	7.3	10.3	9.0	7.9	5.4	6.2	6.3	9.7	
	5% WB	MCDBR	9.2	10.4	9.1	7.9	9.7	15.5	12.6	9.2	6.4	6.5	6.5	9.9	
		MCWBR	9.5	10.7	8.9	7.3	7.1	10.0	8.5	6.8	4.7	6.2	6.7	9.7	
Clear-Sky Solar Irradiance	taub	0.176	0.256	0.285	0.346	0.381	0.370	0.375	0.387	0.329	0.292	0.392	N/A		
	taud	1.934	2.117	2.146	2.024	2.090	2.381	2.427	2.342	2.470	2.289	2.639	N/A		
	Ebn at Noon	127	206	252	257	259	268	263	246	241	192	90	N/A		
	Edh at Noon	8	19	29	42	43	33	31	31	22	16	8	N/A		
All-Sky Solar Radiation	RadAvg	45	253	794	1363	1525	1722	1411	1033	672	257	62	16		
	RadStd	5	41	76	105	175	126	161	112	76	37	9	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page